DO NOW — RECALL

Autumn mock feedback and intervention

Name: _____ Class: _____ Date: _____

1. Why must binary search use sorted input?

2. What is a base case?

3. Why do exam questions award method marks?

4. A rising-edge D flip-flop samples D=1 at the first clock edge. D then changes to 0 between edges, and another rising edge occurs. State Q immediately after the first edge, between edges, and after the second edge.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It discards half the candidates based on ordering.

2. Model answer

A condition that stops recursive calls.

3. Model answer

They credit correct intermediate reasoning and algorithm states.

4. Model answer

Q=1 immediately after first rising edge; remains 1 between edges even though D changes; at next rising edge Q becomes 0.

Vocabulary — know these terms

diagnostic — Assessment used to locate a learning gap.

method mark — Credit for a correct intermediate step.

transfer — Applying a method to a changed problem.

correction — Justified replacement for incorrect reasoning.

NEW LEARNING

Read, annotate and explain

Diagnose the weakness

Look for a pattern across the marked questions: omitted intermediate states, incorrect notation or missed explanations.

Rebuild the method

Model each calculation, comparison or state change instead of simply copying the final answer.

Test transfer

Give students a different array or bit pattern and check whether they can produce the same method independently.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Correct an incomplete two's-complement answer for −9 in eight bits.

Predict first

Start with +9 in exactly eight bits and invert it.

Explain the answer

Add one to give 11110111, representing −9. This explicitly shows the stages.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Improve an answer to 'trace linear search for 7 in [3,5,7,9]' by showing comparisons and returned index.

Question 2

Correct the claim '−6 is 11111001 in eight-bit two's complement'. Provide the full method.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A student traces binary search for 13 in [1,3,5,7,9,11,13] but does not change high or low after a comparison. Rewrite the trace with bounds and explain why it terminates.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Copying only a mark-scheme answer proves you can repeat the method.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.